

Wind-Assisted Propulsion on North Sea Routes

Maritime Performance Analytics for Flettner Rotors

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What this presentation covers

I Background & Data

Context, Magnus effect, vessel spec, AIS + metocean dataset

II Methodology & Models

Feature engineering, model hierarchy, speed prediction results

III Rotor Performance

Speed uplift, weather windows, gain surface, energy & CO₂

IV Route Intelligence

Route segmentation, Dijkstra optimisation, high-gain detection

V Uncertainty & Conclusions

Monte Carlo routing, key findings, operational takeaways

Part I

Background & Data

Project context · Vessel specification · Dataset

Project context: why Flettner rotors?

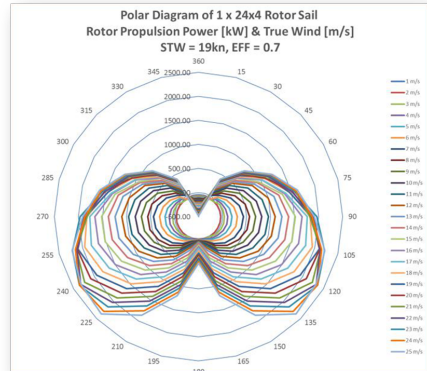
The Magnus effect

- ▶ A rotating cylinder in an airflow generates a transverse lift force.
- ▶ If part of that force aligns with the vessel heading, it reduces required engine power.
- ▶ A single rotor can contribute 40-300 kW depending on wind speed and angle.

Why now?

- ▶ IMO targets 50% GHG reduction by 2050 (Fourth IMO GHG Study 2020).
- ▶ Fuel at \$600-900/t; even 1% savings matters at scale.
- ▶ Rotors are retrofittable; no propulsion redesign required.

Challenge: benefit is *conditional* on wind speed, angle, sea state, and vessel heading - not a constant uplift



Vessel and rotor specification

STANDARD PROFILE (No Deck Cargo)

NOMENCLATURE:

PROF = Profile curve
COEF = Coefficient
AREA1 = AREA blw WL m2
AREA2 = AREA abv WL m2
ZCG1 = Vert.CoG AREA1 abv BL m
ZCG2 = Vert.CoG AREA2 abv BL m
LEVER = Distance between Vert.CoG (ZCG2-ZCG1) m
MOM = Moment from center of underwater part m3

PROFILE

Loading condition: T=7.099, TR=0 Draught= 7.099 m Trim= 0.000 m

PROF	COEF	AREA2 m2	ZCG2 m	AREA1 m2	ZCG1 m	LEVER m	MOM m3
PROFILE	1.000	2339.9	18.790	1034.5	3.590	15.200	35567.6

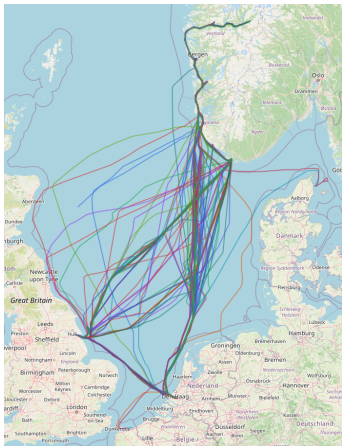
Key parameters used in the analysis

- ▶ Design speed: 11.5 kn
- ▶ Design brake power: 3,500 kW
- ▶ Resistance model: cubic (power $\propto$ speed³)
- ▶ SFOC: 195 g/kWh
- ▶ HFO CO₂ factor: 3.114 t CO₂/t fuel
- ▶ Rotor active conditions: $P_{\text{rotor}} > 0$,
 $H_s \leq 6$ m, $U_{\text{wind}} \leq 42$ m/s
- ▶ Power-to-speed conversion: 200 kW $\approx$ 1 kn at operating point

Note

All scenario values flow from a single set of consistent assumptions applied throughout every extension notebook.

Data sources and trajectories



North Sea AIS
traces on OpenStreetMap basemap.

Property	Value
Period	2023-02-02 to 2026-02-02
Voyages	528
Total rows	125,160
Training voyages	422 (88,346 rows)
Test voyages	106 (35,758 rows)

Data layers joined per AIS point

- ▶ **ERA5:** 10 m wind components, gusts
- ▶ **Copernicus Marine:** significant wave height H_s , mean wave direction & period
- ▶ **Copernicus currents:** eastward & northward current components

Key

Spatio-temporal interpolation attaches

Part II

Methodology & Models

Feature engineering · Model hierarchy · Speed prediction results

Feature engineering: leakage-aware design

Feature groups

- ▶ **Wind/wave:** true wind speed, H_s , wave period, wind^2 , wave-wind alignment
- ▶ **Angle encoding:** $\sin \alpha$, $\cos \alpha$ for relative wind - avoids 0/360 discontinuity
- ▶ **Apparent wind:** speed & angle as experienced by the moving vessel
- ▶ **Currents:** magnitude, direction, along-course and cross-course components
- ▶ **Nowcast lags:** SOG_{t-1} , SOG_{t-2} , rolling mean_3 - grouped by voyage
- ▶ **Interactions:** wind-angle, wave-wind, current-course nonlinear terms

Leakage controls

- ▶ Lag features never use the *current* target row
- ▶ Rolling windows reset at voyage boundaries - no cross-voyage borrowing
- ▶ Rotor variables are **not** training features; applied post-prediction only
- ▶ Chronological voyage split: test set is later in time than training set

Key takeaway

Weather-only and lagged nowcast models use *disjoint* feature sets - explanatory and predictive questions are kept separate.

Model hierarchy and rotor scenario methodology

Four model levels for SOG prediction

Model	Description
B1	Constant mean training speed
B2	Linear: wind speed, H_s , $\sin \alpha$, $\cos \alpha$
M1	XGBoost, weather-only features
M2	XGBoost, weather + lagged SOG (nowcast)

XGBoost: 300 trees, depth 6, lr 0.05, 80% row/col subsampling.

95% prediction interval (M2 quantile regression): mean width 3.161 kn, empirical coverage 96.5%.

Rotor scenario (post-prediction layer)

$$\Delta \text{SOG} = \frac{P_{\text{rotor}}}{200 \text{ kW/kn}}$$

Active when $P_{\text{rotor}} > 0$, $H_s \leq 6 \text{ m}$, $U \leq 42 \text{ m/s}$.
 P_{rotor} read from the supplied polar-response table (wind speed $\times$ relative angle).

Weather window criterion

$$\text{SOG} \geq 10 \text{ kn} \quad \text{and} \quad H_s \leq 3 \text{ m}$$

Contiguous observations satisfying both = one window.

Design choice

Rotor is a scenario overlay, not a training feature - baseline and rotor-assisted results are directly comparable.

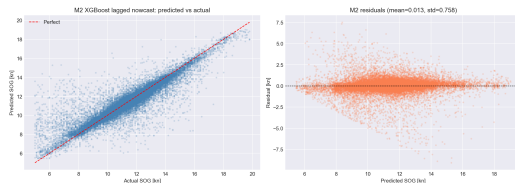
Model results: held-out test performance

Model	MAE	RMSE	R^2	$\pm 1\text{kn}$
B1: Constant	1.388	1.893	-0.002	49.3%
B2: Linear	1.377	1.860	0.034	49.5%
M1: XGB weather	1.468	1.883	0.010	41.7%
M2: Nowcast	0.406	0.758	0.839	91.7%

All errors in knots. Voyage split: 106 test voyages.

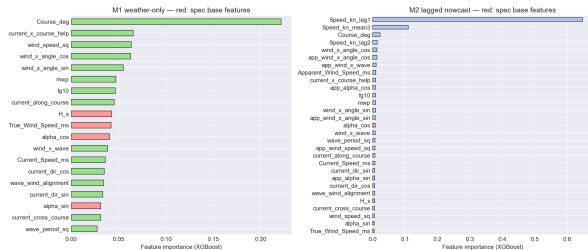
Interpretation

- ▶ B1/B2 nearly identical - wind/wave has limited linear signal.
- ▶ M1 does not beat linear baseline - weather-only generalisation is hard.
- ▶ M2 works: recent SOG encodes engine setting, cargo, operational state.



M2 predicted vs. actual SOG (test set).

Feature importance and prediction intervals



Weather-only (left) and lagged nowcast (right)
XGBoost importances.

Weather-only (M1): course and current interaction terms rank highest.

Wind/wave present but not dominant - missing operational variables limit explanatory power.

Nowcast (M2): lagged SOG dominates - recent speed summarises engine setting, route, and loading. Apparent wind angle enters top features; weather secondary.

Key takeaway

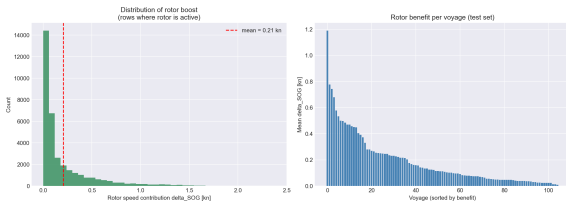
Speed persistence is the main signal; weather effects need additional operational variables to isolate cleanly.

Part III

Rotor Performance

Speed uplift · Weather windows · Gain surface · Energy & CO₂

Rotor scenario: speed uplift and weather-window impact



Rotor power distribution and per-voyage mean speed contribution.

Test-set rotor summary

Active share	98.1%
Mean active power	41.2 kW
Mean active Δ SOG	0.206 kn
Overall mean Δ SOG	0.202 kn

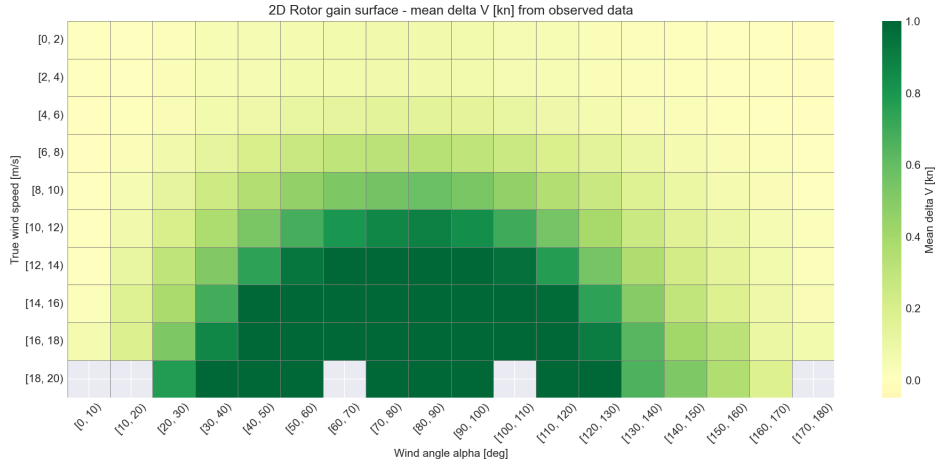
Weather-window impact

	Windows	Coverage
Baseline	866	76.0%
+Rotor	808	77.5%
Δ	-58	+1.5 pp

Key takeaway

Fewer but longer windows: rotor merges marginal periods. Coverage +529 test points inside windows.

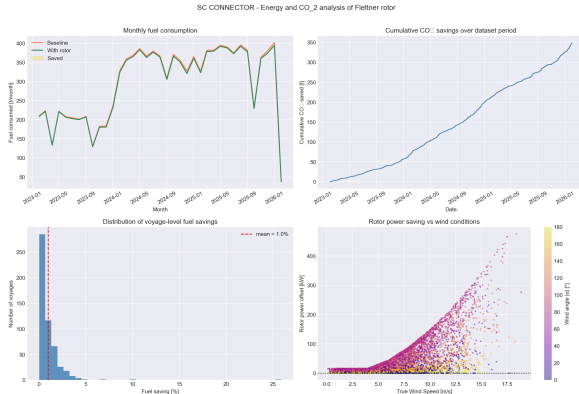
Rotor gain surface: where does it help?



Gain concentrated at beam/quartering angles ($50\text{--}130^\circ$) above 8 m/s. Headwind/tailwind sectors poor. Sea state acts as a gate, not a gain driver.

Key takeaway

Extension 1: Energy, fuel, and CO₂ impact



Fuel and CO₂ impact from the rotor energy model.

Three-year dataset summary

Rotor active share	97.6%
Fuel baseline	11,067 t
Fuel w/ rotor	10,954 t
Fuel saved	112.2 t (1.01%)
CO ₂ saved	349.4 t (1.01%)

Annualised estimate

- ▶ 37.4 t fuel saved/year
- ▶ 116.4 t CO₂ avoided/year
- ▶ CII: 53.75 → 53.20 g CO₂/(t nm)

Note

Cubic resistance model; scenario estimate, not certified trial. Order of magnitude is the robust finding.

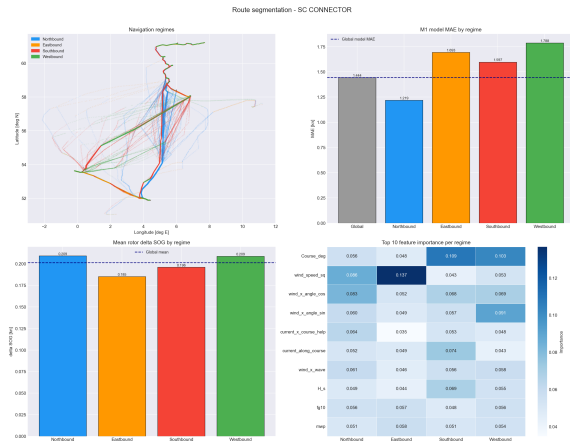
Part IV

Route Intelligence

Route segmentation · Dijkstra optimisation · High-gain detection

Extension 2: Route segmentation by heading regime

Regime summary (full dataset)



Route segmentation into N/E/S/W heading regimes.

Regime	Mean SOG	Mean Δ SOG
Northbound	12.05 kn	0.209 kn
Eastbound	12.34 kn	0.185 kn
Southbound	10.95 kn	0.196 kn
Westbound	10.58 kn	0.209 kn

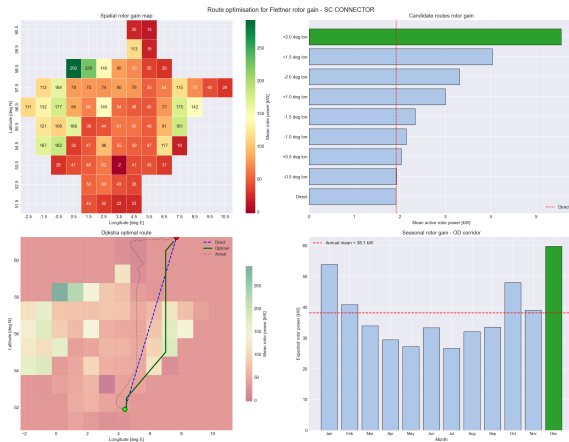
Weather-window coverage change

- ▶ Westbound: 85.1% → **90.1%** (largest gain)
- ▶ Southbound: 82.3% → 84.5%
- ▶ N/E already at high baseline - smaller delta

Key takeaway

Westbound legs benefit most - same vessel, different heading geometry creates different

Extension 3: Route optimisation - Dijkstra proof of concept



Dijkstra grid-search routing for rotor-gain optimisation (Voyage 456).

Voyage 456 result (Rotterdam → W. Norway)

	Direct	Optimal
Distance	571 nm	568 nm
Mean power	1.9 kW	7.5 kW
Δ SOG	0.010 kn	0.037 kn
Gain		+288% relative

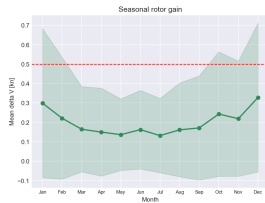
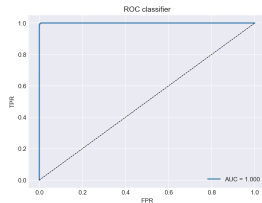
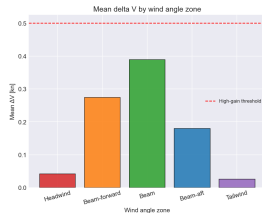
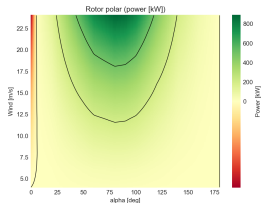
Seasonal best months: December (59.8 kW avg) vs. July (26.6 kW avg)

Note

Proof of method: grid is coarse, built from historical fields. Production use needs forecast fields, land mask, traffic separation, safety rules.

Extension 4: High-gain moment detection

Rotor gain detection - summary



Gain

detection: high-gain ($\Delta\text{SOG} \geq 0.5$ kn) moment distribution.

High-gain moments ($\Delta\text{SOG} \geq 0.5$ kn): 14,856 obs. = **11.9%** of dataset. Median wind 11.5 m/s; median angle 82° (80% between 50 - 113°).

XGBoost classifier

Accuracy	99.8%
Precision	99.4%
Recall	99.0%
F1	0.992
ROC-AUC	1.000

Transparent rule: wind ≥ 8 m/s & angle 50 - 130° gives recall 95.8%, accuracy 94.2%. Easy to apply to forecasts on the bridge.

Key takeaway

Only 12% of moments are high-gain, but they carry the bulk of rotor value - plan around them.

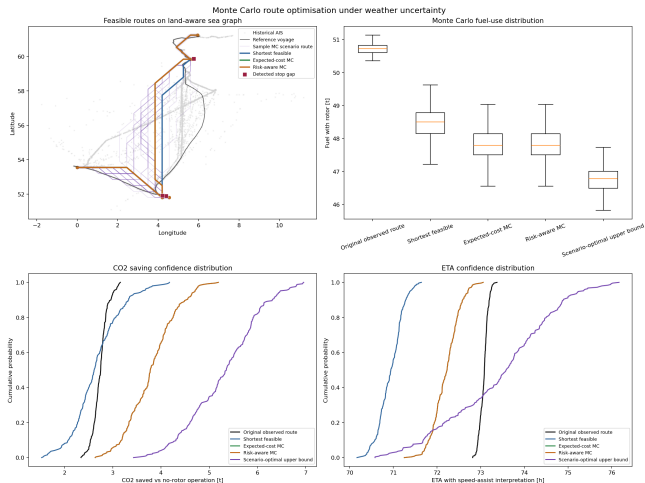
Part V

Uncertainty & Conclusions

Monte Carlo routing · Key findings · Operational takeaways

Extension 5: Monte Carlo routing under uncertainty

Voyage 364 - four route variants
(land-masked graph: 537 nodes,
3,784 edges)



Route	Fuel (t)	Saved (t)
Observed	50.72	0.88
Shortest feasible	48.47	0.86
MC expected-cost	47.81	1.22
Scenario upper bd	46.75	1.70

- ▶ MC route beats shortest-feasible with **85.6% probability**
- ▶ CO₂ saved: up to 5.3 t/voyage (upper bound)
- ▶ 4 itinerary legs, 3 detected port stops

Key takeaway

Uncertainty-aware routing

Monte Carlo routing: 1,000 scenario samples for Voyage
364

Conclusion

Modelling

- ▶ Lagged nowcast XGBoost: MAE **0.406 kn**, R^2 **0.839** - strong short-horizon SOG prediction.
- ▶ Weather-only models are limited by missing operational variables (engine, draught, loading).
- ▶ Chronological voyage split prevents temporal leakage and tests real generalisation.

Rotor scenario

- ▶ Mean speed uplift **0.202 kn**; weather-window coverage **+1.5 pp**.
- ▶ Only **11.9%** of moments are high-gain - rotor is a conditional asset.
- ▶ Rule: wind ≥ 8 m/s at beam/quartering angle (50-130°).

Operational impact (scenario estimates)

- ▶ 37.4 t fuel / 116 t CO₂ saved per year.
- ▶ CII improvement 53.75 → 53.20 g CO₂/(t nm).
- ▶ Westbound legs gain most from rotor assistance.

What was built

- ▶ Reproducible pipeline: AIS + ERA5 + Copernicus Marine.
- ▶ Five extensions: energy, segmentation, routing, gain detection, Monte Carlo.
- ▶ Transparent assumptions throughout - auditable scenario, not a black-box claim.

Final takeaway

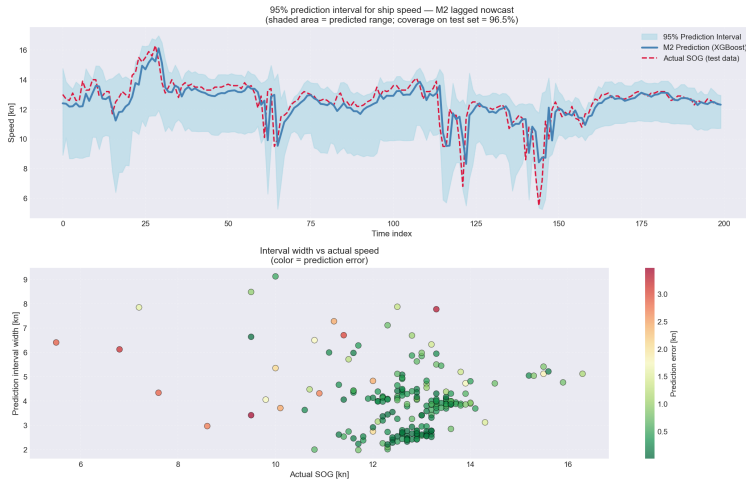
Wind-assisted propulsion is an opportunity map, not a constant bonus. The project identifies

Thank you!

Questions welcome.

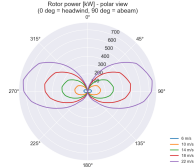
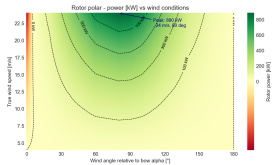
Detailed methodology, assumptions, and extension figures follow in the appendix.

Appendix: prediction intervals (95%)

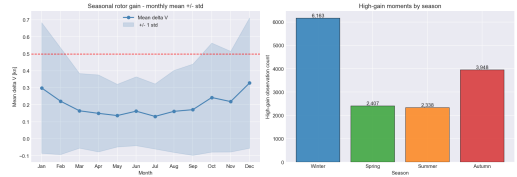


95% quantile-regression prediction intervals for the nowcast model. Mean width 3.161 kn; empirical coverage 96.5% on the test set - slightly conservative but well calibrated.

Appendix: rotor polar and seasonal variation

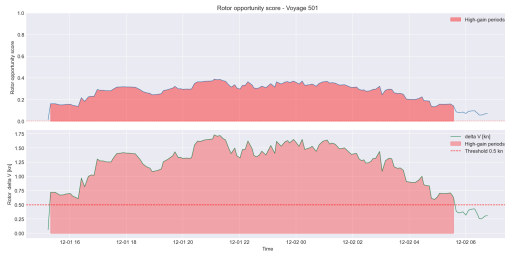


Rotor polar gain diagram.

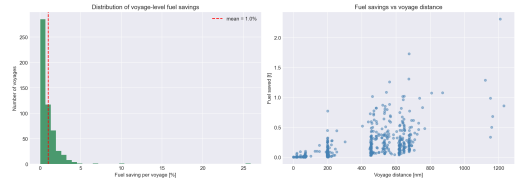


Seasonal rotor gain variation. December best (59.8 kW), July worst (26.6 kW). Winter wind is stronger but wave risk is higher.

Appendix: rotor opportunity by voyage

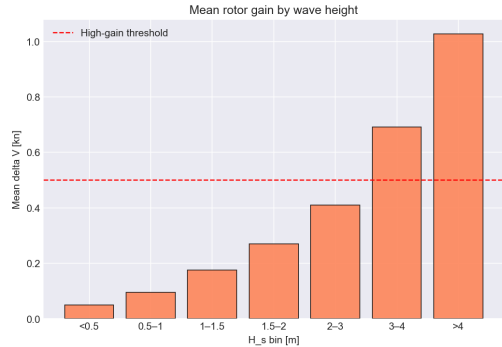
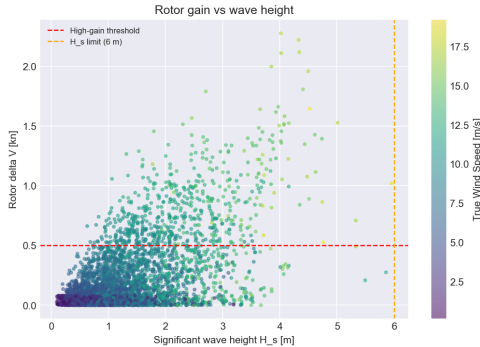


Voyage 501: rotor opportunity map along the track.



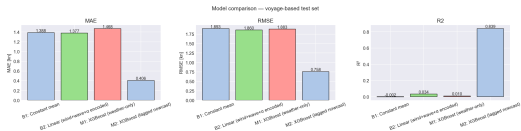
Per-voyage fuel savings distribution.

Appendix: rotor wave-height interaction

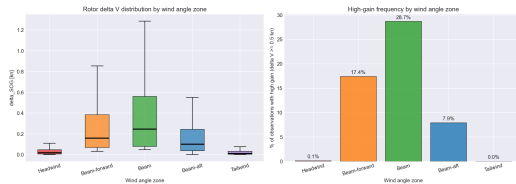


Wave height interaction with rotor gain. H_s primarily acts as an operational gate (rotor off above 6 m); it is not the main driver of gain magnitude. Wind angle and wind speed create the opportunity; sea state can remove it.

Appendix: model comparison and zone analysis



All four model MAE/RMSE/ R^2 side-by-side.



Rotor zone analysis across the study area.

Appendix: rotor scenario methodology - full formula

Rotor power from polar table:

$$P_{\text{rotor}} = f_{\text{polar}}(U_{\text{wind}}, |\alpha|)$$

where $|\alpha|$ is the absolute relative wind angle (0 = headwind, 90 = beam).

Active condition:

$$\text{active} = \mathbf{1}[P_{\text{rotor}} > 0] \cdot \mathbf{1}[H_s \leq 6 \text{ m}] \cdot \mathbf{1}[U_{\text{wind}} \leq 42 \text{ m/s}]$$

Speed contribution:

$$\Delta\text{SOG} = \text{active} \times \frac{P_{\text{rotor}}}{200 \text{ kW/kn}}$$

Fuel saving rate (cubic resistance model):

$$\dot{m}_{\text{fuel}} = \frac{P_{\text{brake}} \cdot \text{SFOC}}{1000} \quad \text{where} \quad P_{\text{brake}} = P_{\text{design}} \left(\frac{\text{SOG}}{v_{\text{design}}} \right)^3 - P_{\text{rotor}}$$

Design principle

Rotor variables are a *scenario overlay* applied after prediction - the baseline model trains on observed operations only, keeping prediction and scenario logics cleanly separated.